

Exp no:9 Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

Aim: To write program for Dielectric - free space boundary condition in Electric field using SCILAB.

Software required:

- SCILAB version 6.0.1

⑨ Given

$$\theta_1 = 30^\circ$$

$$\theta_2 = 60^\circ$$

$$E_{r1} = 4$$

$$\tan \theta_1 = \tan 30^\circ = 0.5774$$

$$\tan \theta_2 = \tan 60^\circ = 1.7321$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{0.5774}{1.7321} = 0.3333$$

$$E_{r2} = \frac{E_{r1}}{0.3333} = \frac{4}{0.3333} = 12.0000$$

$$E_{r2} = 12.000$$

Case 1: In this case, the dielectric medium has a relative permittivity (ϵ_r) of 2, meaning the dielectric reduces the electric field by half compared to the free space. This is typical for materials like mica or certain types of plastics.

▣ Free Space Electric Field (E_1): 100 V/m

▣ Dielectric Electric Field (E_2): 50 V/m

Case 2: Here, the dielectric medium has a higher relative permittivity ($\epsilon_r=2.5$) compared to Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like glass or water.

▣ Parameters:

o $E_{n1} = 200$ V/m (Electric field in free space).

o $E_{2n} = 80$ V/m (Electric field in the dielectric medium).

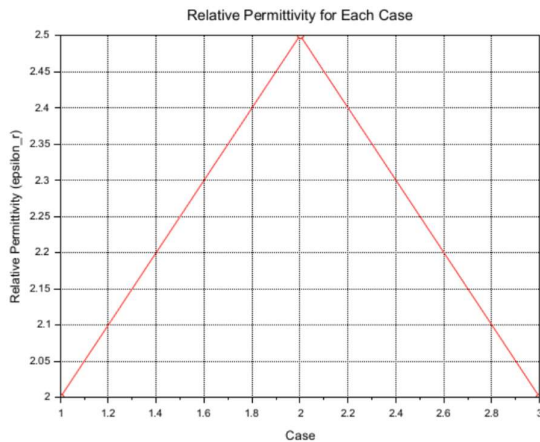
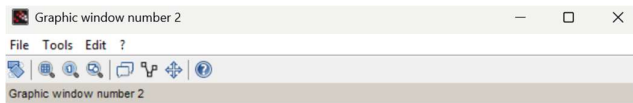
Case 3: The dielectric medium here also has a relative permittivity of 2. The behavior of the material suggests it could be another low-polarization dielectric material, with a consistent reduction of the electric field by half.

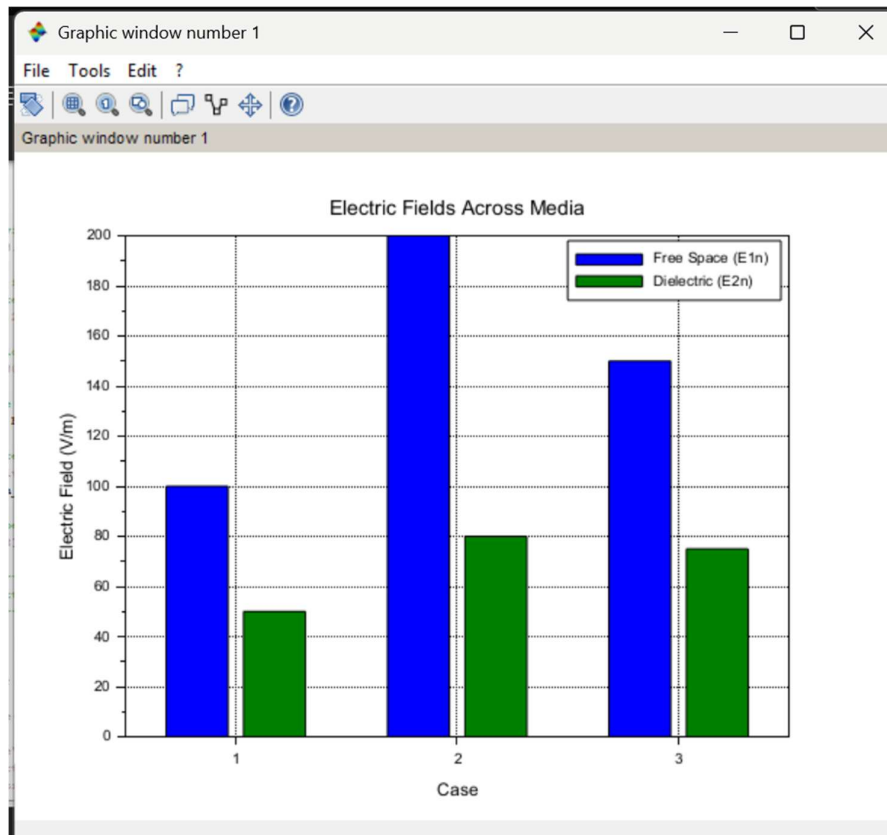
▣ Free Space Electric Field (E_{1n}): 150 V/m

▣ Dielectric Electric Field (E_{2n}): 75 V/m

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9.1.sci (C:\Users\abhis\Documents\9.1.sci) - SciNotes
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1 clc;
2 clear;
3 clf;
4
5 //Permittivity of free space
6 epsilon0 = 8.854e-12;
7
8 //Electric field values
9 //Free space (V/m)
10 E1n = [100, -200, -150];
11
12 //Dielectric (V/m)
13 E2n = [50, -80, -75];
14
15 //Calculate relative permittivity
16 epsilon_r = E1n ./ E2n;
17
18 //Display results
19 disp("Relative Permittivity for Each Case:");
20 disp(epsilon_r);
21
22 //Case numbers
23 x = [1, -2, -3];
24
25 //-----
26 //Plot Electric Fields
27 //-----
28 scf(1);
29 clf();
30
31 bar(x, [E1n; E2n]');
32
33 legend("Free Space (E1n)", "Dielectric (E2n)");
34
35 xlabel("Case");
36 ylabel("Electric Field (V/m)");
37 title("Electric Fields Across Media");
```





Result : Thus the program was verified for Dielectric - free space boundary condition in Electric field using SCILAB was successfully.